Cost of Living Analysis

Earliest Date
8/1/1990

Quarter
Q1

Clear all slicers

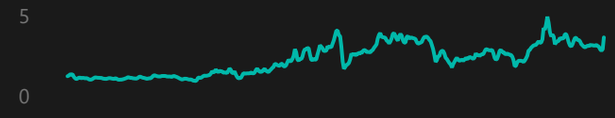
Real Wage Index



Consumer Price Index

212.15

Average Regular Gas Price

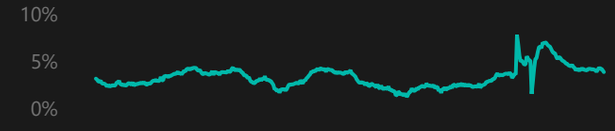


Average Gas Price

$2.25



Hourly Wage Growth



Wage Growth

3.27%



CPI Inflation



Inflation

2.59%



Year	Quarter	Month
☐ 1990	☐ Q1	☐ January
☐ 1991	☐ Q2	☐ February
☐ 1992	☐ Q3	☐ March
☐ 1993	☐ Q4	☐ April
☐ 1994		☐ May
☐ 1995		☐ June
☐ 1996		☐ July
☐ 1997		☐ August

Key Findings

- Current Summary
  - We use four indicators to make a analyzes the state of cost of living in the United States since 1990. The main indicator used in this dashboard is the Consumer Price Index (CPI), used as a foundations for the other indicators to be compared to.
  - The use of average Gas Price is to show how a singular component of CPI can have a different trend.
  - Hourly Wage Growth is an economical indicator that affect all working people that is a component of buying power.
  - By taking the incremental changes between years in CPI we can than use the difference to graph the percentage change every year to see percentage change on the prices of general services and goods.

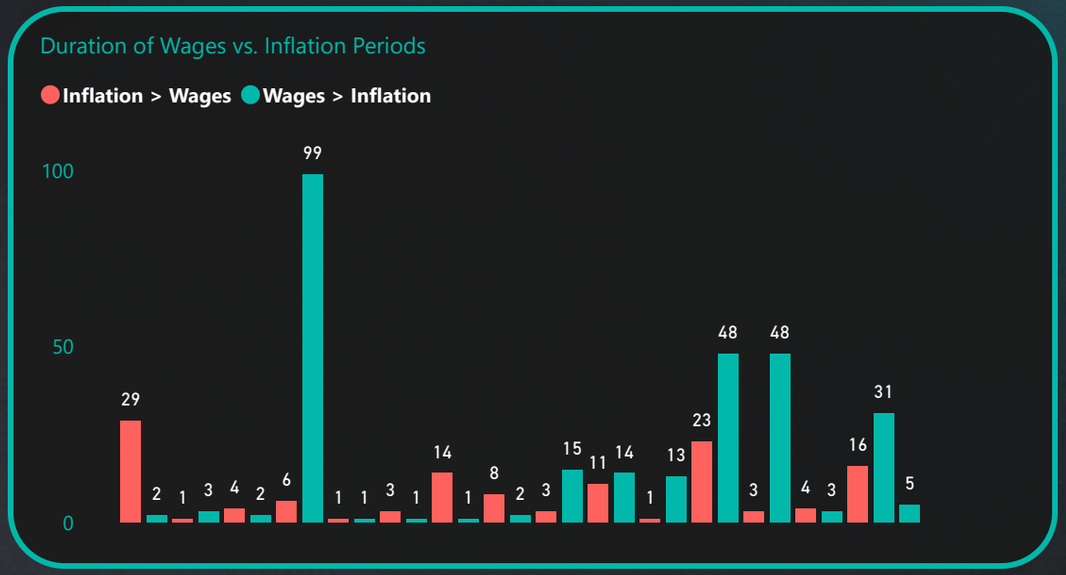
- Key Observation
  - Consumer Pride Index (CPI) Has seen steady growth in general over the years, with observations of spikes being between 2008 - 2011 & 2020 - 2023. By using Inflation we can quickly extract this information for further analysis.
  - There visual appears to be no visual correlations with these two events with average gas prices. However, Hourly Wage Growth may have a strong correlations that will need further analysis by comparing with other indicators.
- Recommendation
- Calculate and find any correlations of the change of inflation with any major evens in the United States and see how that may effect one the indicators we used to find strong evidence of it affecting buying power.

Wage Comparison



Latest CPI YoY %	Latest Wage YoY %	Latest Real Wage Growth
3.3%	**3.82%**	**0.50%**





Summary:

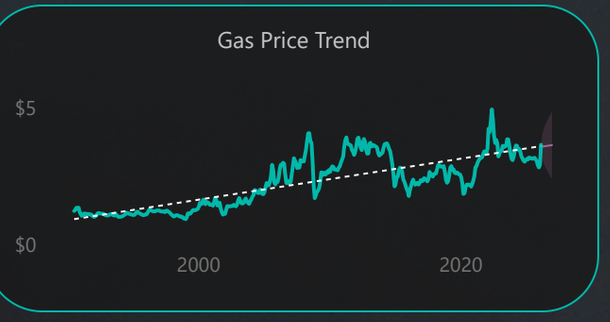
- Observation
  - There is a recurring cycle between periods where wage growth exceeds inflation and periods where inflation exceeds wage growth.
  - These positive and negative Real Wage Gap periods vary considerably in duration.

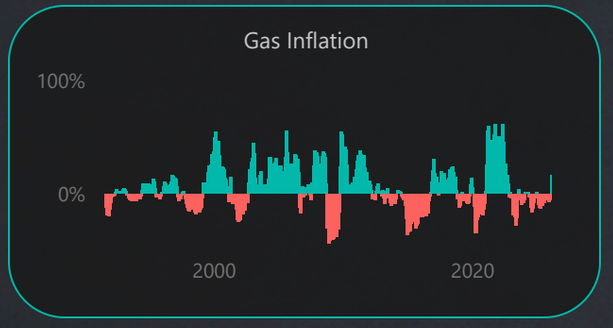
- Interpretation
  - The duration of these periods ranges from as little as 1 month to as long as 99 months (8 years and 3 months).
  - Periods where inflation outpaced wage growth include several extended runs of 29, 23, 16, and 14 months.
  - However, some periods where wage growth outpaced inflation lasted substantially longer, including runs of 99, 48, 31, and 15 months.
  - This suggests that wage growth can remain above inflation for extended periods, but the duration of both positive and negative Real Wage Gap periods varies considerably.

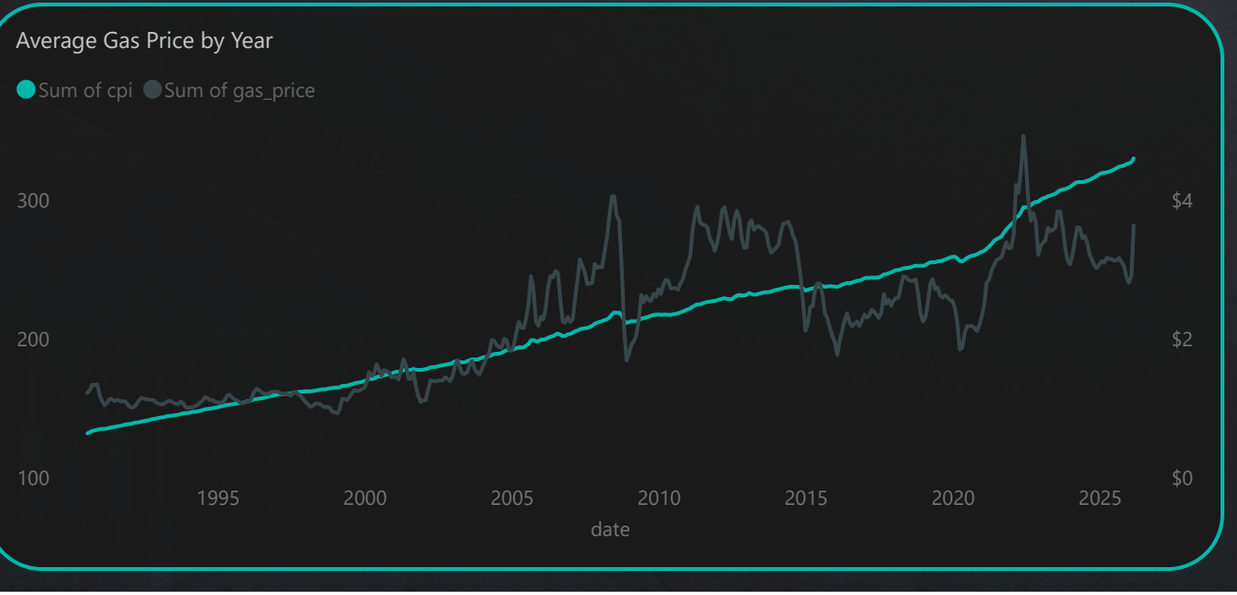
- Why it Matters
  - Inflation and wage growth have opposing effects on purchasing power: inflation increases the cost of goods and services, while wage growth increases the income available to purchase them.
  - The changing gap between the two helps show when workers' purchasing power is under pressure and when wages are gaining ground relative to prices.
  - While this analysis is not sufficient to predict future cycles, the historical data shows several notably long

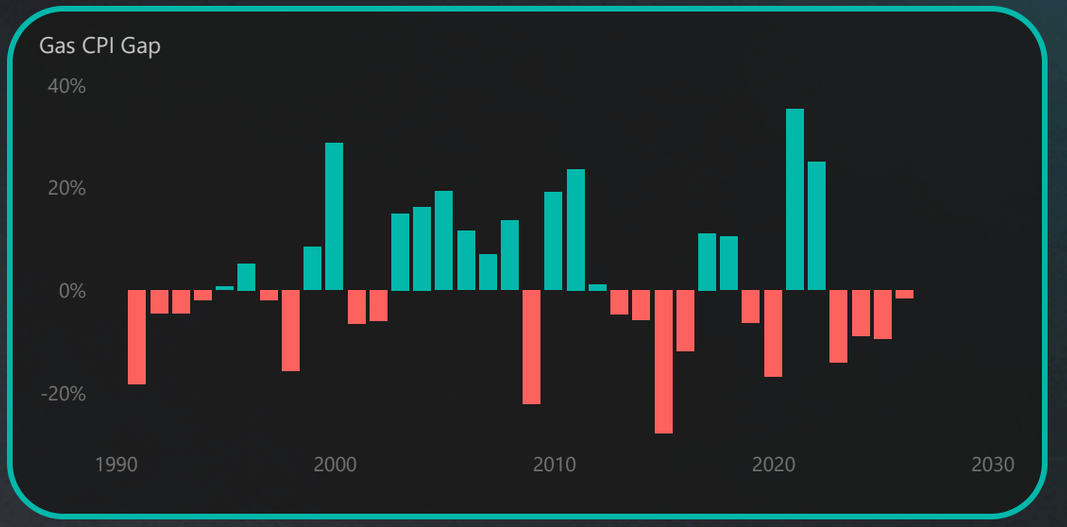
Gas Prices vs. Inflation

Latest Gas Price	Highest Gas Price	Lowest Gas Price
$3.64	**$4.929**	**$0.921**









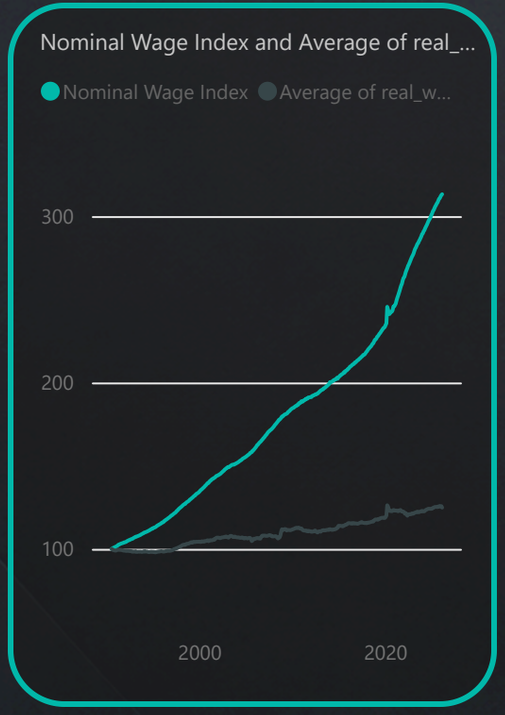
Summary:

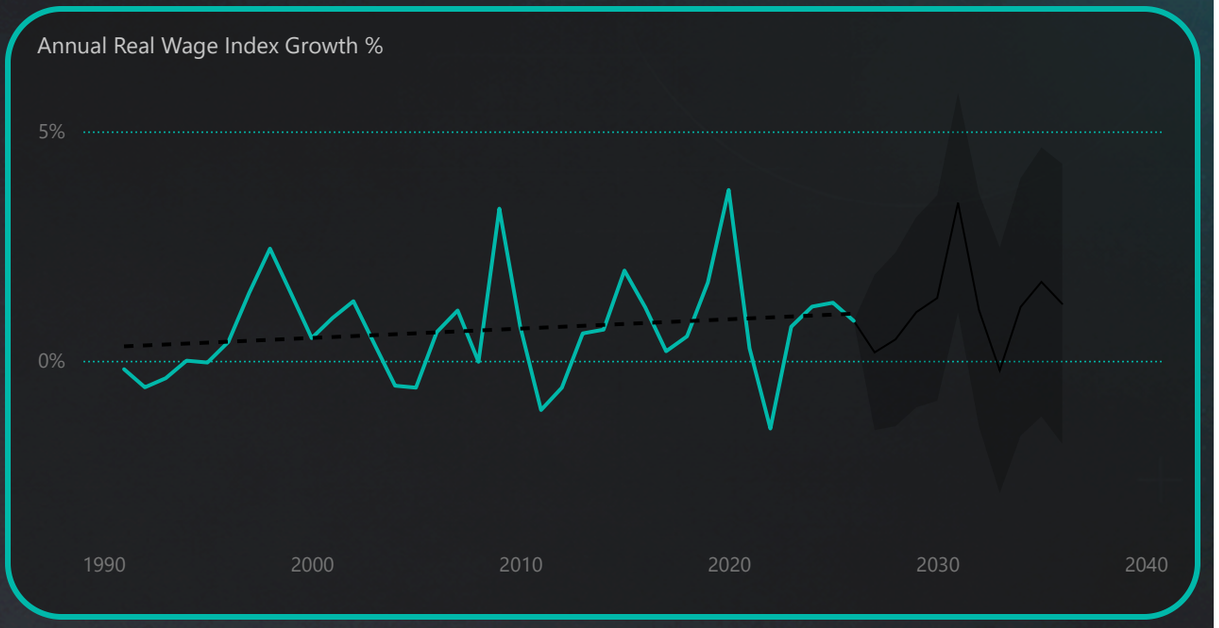
- Observation
- U.S. gasoline prices show substantial short-term volatility but a long-term upward trend.
- Gasoline inflation fluctuates between positive and negative growth, with several periods of sharp price changes.
- Gasoline price growth frequently differs from overall CPI inflation, indicating that fuel prices do not consistently follow broader consumer price trends.
- Interpretation
- Although gasoline prices generally increase over the long term, their growth is inconsistent, with periods of rapid increases followed by declines or stabilization.
- The differences between gasoline inflation and overall CPI inflation suggest that gasoline prices are influenced by factors beyond general inflation.
- Unlike the relatively steady upward movement of CPI, gasoline prices experience more pronounced fluctuations.
- Why it Matters
- Gasoline price volatility can create unpredictable transportation expenses for consumers and businesses.
- When gasoline inflation exceeds overall CPI inflation, fuel-dependent households may experience greater cost pressures than the broader inflation rate suggests.
- Understanding these differences helps explain why changes in the cost of living can vary across

Wage

Latest Real Wage ...	Latest Wage YoY %	Latest CPI YoY %
124.8	3.82%	3.3%







Summary:

- Observation
- Real wage growth fluctuates between positive and negative periods, indicating that wage increases do not consistently outpace inflation.
- The duration and magnitude of these periods vary, with some years experiencing substantial gains or losses in purchasing power.
- Despite these fluctuations, the real wage index demonstrates an overall upward trend over the historical period.

- Interpretation
- Positive real wage growth indicates that wages are increasing faster than consumer prices, improving the purchasing power of wages. Negative growth indicates the opposite.
- The long-term increase in the real wage index suggests that inflation-adjusted wages have improved overall, despite temporary declines.
- Nominal wages have increased substantially faster than real wages, demonstrating how inflation reduces the purchasing-power gains associated with wage increases.

- Why it Matters
- Nominal wage growth alone does not accurately represent improvements in workers' purchasing power.
- Periods of negative real wage growth can place financial pressure on households, even when their nominal earnings are increasing.
- Comparing nominal and real wage trends provides a more accurate assessment of changes in workers' economic well-being over